

UIL Official List of Boolean Algebra Identities (Laws)

- 1	$A + A = A$	Idempotent Law for OR
- 2	$A * A = A$	Idempotent Law for AND
- 3	$A + B = B + A$	Commutative Law for OR
- 4	$A * B = B * A$	Commutative Law for AND
- 5	$A + (B + C) = (A + B) + C$	Associative Law for OR
- 6	$A * (B * C) = (A * B) * C$	Associative Law for AND
+ 7	$A * (B + C) = A * B + A * C$	Distributive Law for AND over OR
* 8	$A + B * C = (A + B) * (A + C)$	Distributive Law for OR over AND
- 9	$A + 1 = 1$	Law of Union
- 10	$A * 0 = 0$	Law of Intersection
+ 11	$A * (A + B) = A$	Law of Absorption
+ 12	$A + A * B = A$	Law of Absorption
- 13	$A * 1 = A$	Identity Law for AND
- 14	$A + 0 = A$	Identity Law for OR
- 15	$\overline{\overline{A}} = A$	Double Negative Law
- 16	$A + \overline{A} = 1$	Law of Complement for OR
- 17	$A * \overline{A} = 0$	Law of Complement for AND
+ 18	$\overline{A + B} = \overline{A} * \overline{B}$	DeMorgan's Law
+ 19	$\overline{A * B} = \overline{A} + \overline{B}$	DeMorgan's Law
* 20	$A \oplus B = A * \overline{B} + \overline{A} * B = \overline{A} * B + A * \overline{B}$	Exclusive OR (XOR)
* 21	$\overline{A \oplus B} = A * B + \overline{A} * \overline{B}$	Exclusive NOR (XNOR)
* 22	$A + \overline{A} * B = A + B$	Law of the "disappearing opposite"
* 23	$(A + B) * (A + C) = A + B * C$	Reverse of Law #8
+ 24		
+ 25	$AB + CD = (A + C)(B + C)(A + D)(B + D)$	FOIL (First,Outer,Inner,Last) AND FOIL-ing OR

Note: **AND** will always be expressed explicitly with the * operator

- learn first: 1-6, 9-10, 13-17
- + learn next: 7, 11-12, 18-19, 24
- * learn last: 8, 20-23, 25